Next experiment: complete target-chord decomposition

2026-11-05 Execution-design follow-on to merged PR3438

Status. Proposed construction and finite experiment. No native scientific run, proof build, identity certificate or measured advantage. PCC statement `PCC-20261105-7179e4` has no signed verification. [pcc:unverified]

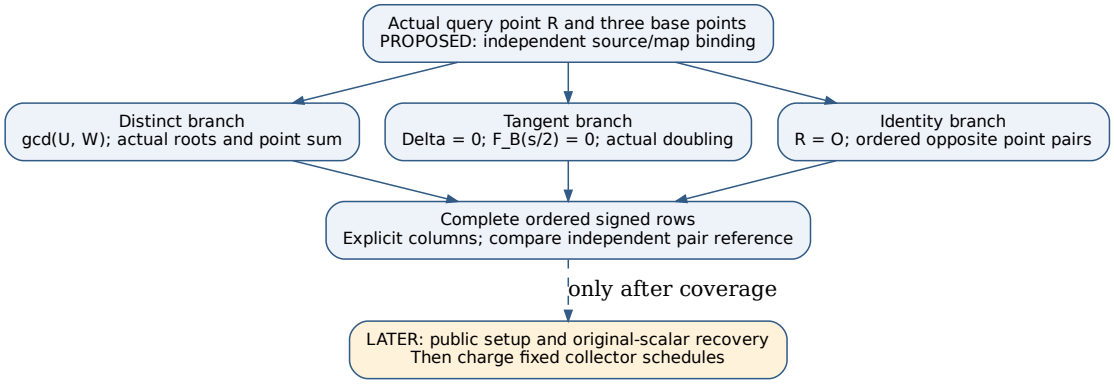
For actual $R = (a, b)$ on $y^2 = x^3 + Ax + B$, the proposed line through $-R$ is $L_l(X) = l(X - a) - b$. Let

$$q_l = X^2 - sX + p, \quad s = l^2 - a, \quad p = al^2 + 2bl + a^2 + A.$$

Proof target: $X^3 + AX + B - L_l(X)^2 = (X - a)q_l(X)$ using the actual target equation. For $F_B = X^3 - c_1X^2 + c_2X - c_3$, prove its remainder is $UX + V$,

$$U = s^2 - p - c_1s + c_2, \quad V = (c_1 - s)p - c_3, \quad W = V + aU.$$

The proposed U, W chart has quartic/cubic degrees, with special leading cancellations recorded honestly. These are proof obligations, not checked degree or runtime results. [pcc:unverified]



The missing branch matters. A repeated-root quadratic need not divide the square-free base polynomial. Keep $\Delta = s^2 - 4p = 0$ and $F_B(s/2) = 0$ for the diagonal branch, with actual point and doubling checks. Handle $R = O$ separately. Prove coverage, including coincident intersections, signs and the nonzero denominator, before algebraic admission. [pcc:unverified]

Background: Semaev, 2004/031; Kosters–Yeo, sections 2 and 4–5. The current atlas already records eliminants, membership and transported-base mechanisms. This design makes no invention claim. [pcc:unverified]

Finite census, then original-source recovery

Actual proposed fixture. $E/\mathbf{F}_{37} : y^2 = x^3 + 7x + 19$, nonsquare $d = 35$ twist $T : y^2 = x^3 + 28x + 33$, $P = (1, 8) \mapsto U = (32, 29)$, subgroup order19. Rebind and recheck historical coordinate Φ and inverse $I = [10]D$ on the actual cyclic subgroups. Relation coefficients use $\mathbf{F}_{19}$, distinct from the curve field. No new registry or generic map claim. [pcc:unverified]

Planned factor	Complete declared population
Base policy	All 84 size-three policies per role
Role	Main, native twist, matched, pulled-back
Query	All 19 subgroup points, including identity
Cells per instrument	$84 \times 4 \times 19 = 6,384$ planned
Nonidentity cells	6,048, repeating 18 public points

These are repeated policy/query/control records, not thousands of distinct DLP challenges. Rust candidate and separate C++ arithmetic reference/auditor are unimplemented. A/A repeats the candidate for deterministic replay. No full-census data exists. [pcc:unverified]

Pass condition. Every required cell completes with zero missing or extra ordered signed rows. Plant distinct, tangent and identity relations through the production path first. False roots, omitted branches, wrong columns/source/map/rank/witness or missing queries must reject changed, rehashed payloads with no receipt. A missed positive control invalidates the instrument; a timeout is incomplete coverage. [pcc:unverified]

Degree/work records. Keep univariate degrees, exact witnesses, construction, extraction, field/group/map and row work separately. A gcd has no Macaulay matrix degree: record null, not zero. Any later matrix comparison uses the same original ideal and expanded original-input witnesses. No intrinsic regularity, equal-weight counter sum or speed result is implied. [pcc:unverified]

Next acceptance. After coverage, freeze separate cold setup and all 18 original-target recoveries per policy/role. Derive public base logs only from admitted relations; retain blocked setups and every failed shift. Check the scalar on original E/P. Then test fixed collector schedules against the entire source-policy frontier, charging both calls, maps and setup. Complete identical per-query row sets cannot yield strict union complementarity. The trace-variable construction remains queued. [pcc:unverified]

Launch dependency. Exact checkpoint-only prefix approval is pending for the new execution branch. The official uploading gate must succeed before any scientific/proof launch. Native/proof/analysis instruments and source hashes are still missing; the canonical protocol is DRAFT. Use at most two existing local workers, 600-second stages and one window no longer than 3600 seconds. No paid resources or arbitrary result uploads.

See the adjacent algorithm, protocol and launch documents.
This diagram has no measured data; old quantitative graphs and benchmarks remain unchanged. [pcc:unverified]